Henry Ford

Mass production, people, and the limits of control



Andy Johnson & Slade McClain
MGT 210, Section T01

The management question

How do you make a complicated product affordable to millions?

Ford connected the product, the production process, and the people doing the work.

Efficiency: fewer resources per car
Effectiveness: cars buyers want and workers can sustain

Management connection: efficiency and effectiveness

The mechanic behind the manager

1863–1947
Raised on a Michigan farm

1903
Ford Motor Company founded

1908
Model T introduced



The fifteen-millionth Ford, a 1927 Model T

[A] [C]

Ford’s influences and collaborators

Direct influences

Edison encouraged Ford’s automotive work.

Meatpacking suggested moving work past fixed stations.

Management context

Smith: division of labor
Taylor: measured, standardized tasks

Ford’s team integrated these principles at scale.

Management connection: division of labor and scientific management

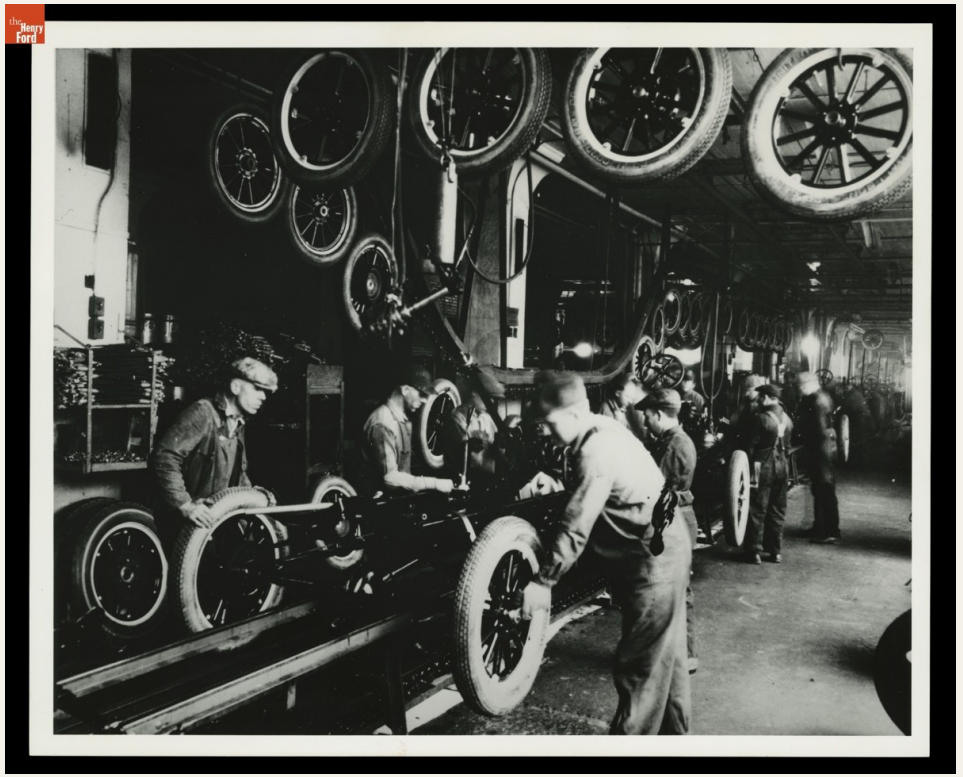
[L] [K]

The production system

Interchangeable parts reduced fitting, rework, and dependence on craft skill.

Narrow jobs shortened training but reduced worker autonomy.

A moving chassis coordinated task order and imposed a common pace.

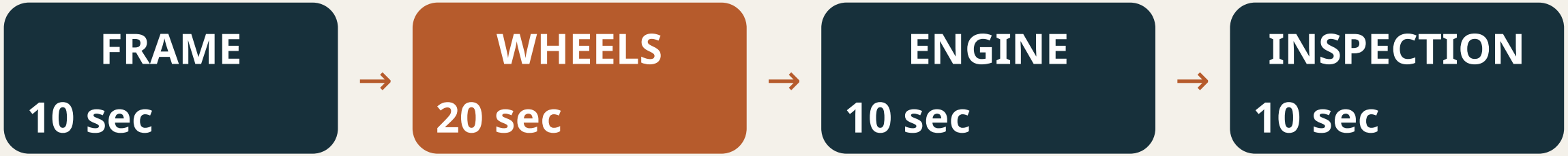


Tire installation at Highland Park, circa 1925

Management connection: work specialization and formalization

[B] [K]

Factory challenge: output or retention?



Teams of 3: operations • worker • finance | One worker per station

A Faster frames	B Add capacity	C Raise pay
10 sec → 5 sec	Second wheels worker	Aim to reduce quitting

Decide: bottleneck, units/minute, and one worker concern.

Then one wheels worker quits. Does your choice still work?

3 minutes • Discuss → staffing shock → vote • Hypothetical classroom model

Ch. 1 efficiency • Ch. 4 decision-making • Ch. 8 job design • Ch. 9 retention

Activity debrief: capacity and retention

Baseline: wheels limits steady-state output to 3 units/min.
A) faster frames still yields 3 units/min.

B) parallel wheels workers can support 6 units/min.
After one quits, capacity returns to 3 units/min.

C) higher pay may reduce quitting. It does not raise the 3-unit physical limit.
A decision needs evidence on cost, retention, quality, and workload.

Ch. 4: evaluate alternatives. Ch. 9: assess retention rather than assume it.

Assembly time fell dramatically

12½ hours **93 minutes**

Approximate Model T chassis assembly time

After the moving-line experiments of 1913–1914

About 88% less elapsed assembly time

93 minutes is assembly duration, not the gap between completed vehicles

[B] [D]

Efficiency expanded the market

$850 **as low as $260**

Model T prices over its production life, in nominal dollars

More than 15 million Model Ts built and sold

A broad market supported a strategy built around low cost.

Management connection: competitive strategy and cost leadership

Fordism as a management system

Reinforcing decisions

A standard product simplifies repeated tasks.

High sales volume spreads factory costs across more units.

The system depends on people

Lower costs can support lower prices and wider demand.

Turnover interrupts the flow. Changing tastes challenge standardization.

Management connection: systems thinking and organizational design

Women at Ford: factory work and pay

WORK

By circa 1913, women made magneto components at Highland Park.

FACTORY LIFE

Specialized, repeatable tasks within a closely managed production system.

PAY ACCESS

Some women qualified for early profit sharing; broader $5 daily minimum in 1916.

Women’s factory employment preceded World War II. Pay access was conditional.

Andy • Job design and compensation shaped the experience of factory life.

[O] The Henry Ford photograph, circa 1913 • [P] Ford & Crowther, ch. VIII • [Q] NYT, 1916

How Ford managed women’s employment

ORGANIZE THE WORK

Standardized tasks

Company-defined jobs and rules

REGULATE ACCESS

Family status shaped eligibility

Ford reported 82 dismissals in 1919

His account says the women were dismissed because their husbands worked.

Later wartime hiring widened roles at Willow Run; it did not prove equal opportunity.

Andy • Ask: can a generous wage coexist with unfair access to work?

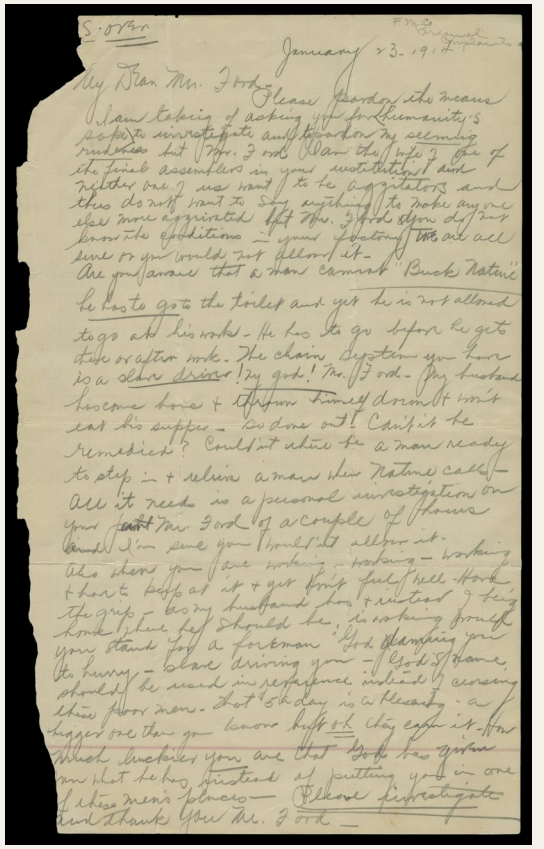
[P] Ford & Crowther, ch. VII–VIII • [R] The Henry Ford, Willow Run • [E] Sociological Department

The staffing problem

Line work reduced autonomy and variety while enforcing a relentless pace.

Workers leaving forced repeated recruitment and training.

Retention became an operating problem as well as an HR problem.



A worker’s wife complained to Ford in 1914

Management connection: retention and job design

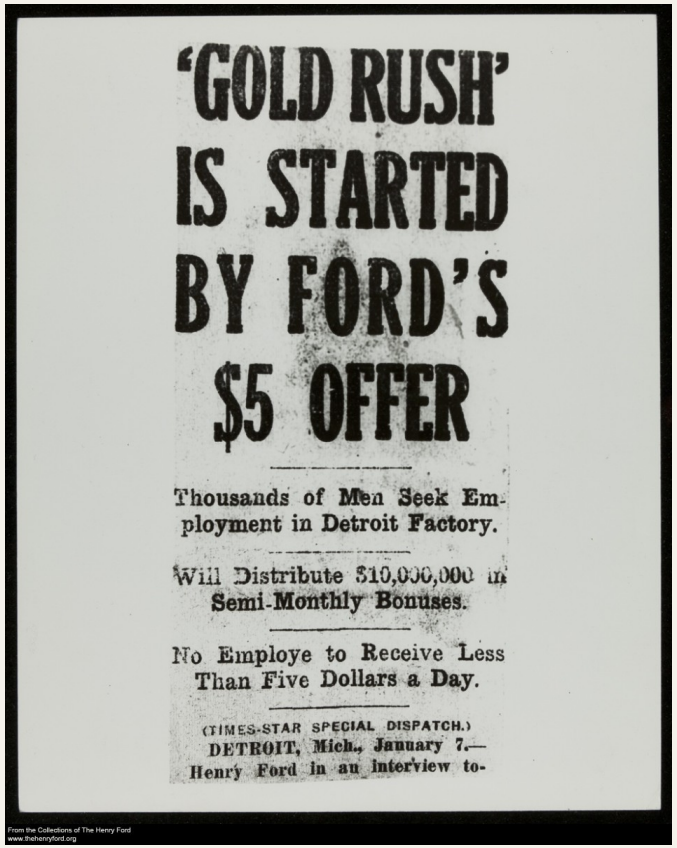
[A] [K]

The $5 day, with conditions

January 5, 1914
Ford and James Couzens announced the plan.

$5 for an 8-hour day
Previous rate: $2.34 for 9 hours

Ordinary wages + conditional profit sharing
Eligibility rules limited access.



'GOLD RUSH' IS STARTED BY FORD'S $5 OFFER

Thousands of Men Seek Employment in Detroit Factory.

Will Distribute $10,000,000 in Semi-Monthly Bonuses.

No Employe to Receive Less Than Five Dollars a Day.

(TIMES-STAR SPECIAL DISPATCH.)
DETROIT, Mich., January 7.—
Henry Ford in an interview to-

From the Collections of The Henry Ford
www.thehenryford.org

Contemporary coverage of the wage announcement

Management connection: compensation and employee retention

[D]

Two labor stories, separated by time

1913–1914

Repetitive line work

High worker turnover

JANUARY 1914

$5 per eight-hour DAY

Conditional pay package

LATER: 1920s–1930s

Bennett’s enforcers

Anti-union intimidation

Supported distinction: pay addressed staffing; later coercion suppressed worker voice.

The sources do not establish “gangsters caused quitting, so Ford raised wages.”

Slade • The later enforcer story does not explain the earlier $5-day decision.

[D] The Henry Ford, “Ford’s Five-Dollar Day” • [N] Detroit Historical Society, “Bennett, Harry” • [I] Battle of the Overpass

When employee welfare became control

Company support

Health and household advice
Financial and legal assistance
English-language education

Company intrusion

Unannounced home visits
Checks on savings and family life
Pay eligibility tied to conduct

Management connection: stakeholder interests, culture, and ethics

[E]

River Rouge and vertical integration

Ford brought more stages of production under company control.

At Rouge: ore docks, steelmaking, glassmaking, and vehicle assembly

The complex achieved “ore to assembly” with the Model A in 1927.

Management connection: vertical integration as a growth strategy

[F]

The innovator resisted change

Buyers wanted more comfort, styling, and choice.

Commitment to the Model T delayed adaptation to the market.

The Model A arrived in 1927 after Ford shut down to retool.



Edsel and Henry Ford with a Model A, 1928

Management connection: environmental scanning and resistance to change

[G]

Labor relations and worker voice

1937: Ford security personnel attacked union organizers at the Battle of the Overpass.

1941: Ford accepted union representation.

High pay did not settle questions of power, dignity, or employee voice.

Management connection: stakeholders and ethical leadership

[I]

Antisemitism and the use of influence

Ford’s newspaper spread antisemitic conspiracy theories beginning in 1920.

The articles circulated internationally as The International Jew.

He issued an apology in 1927. The propaganda’s harm continued.

Management connection: leader accountability and organizational culture

[H] [M]

A balanced evaluation of Ford

Contributions

Mass production expanded access
Product and process choices reinforced each other
Pay helped support staffing

Limits and failures

Control invaded private life
Gender rules restricted opportunity
Rigidity, coercion, and antisemitism

Management connection: efficiency, effectiveness, and ethical responsibility

[C] [D] [E] [G] [H] [I]

Where we still see related ideas

Manufacturing

Standard work and coordinated flow remain central.

Toyota adds pull production and stopping to address defects.

Services and logistics

Kitchens and fulfillment centers also divide work into stations.

These are comparisons of methods, not claims of direct descent.

Management connection: organizational design and continuous improvement

[G] [J]

One company, four management functions

Planning: align affordability with demand, then revisit assumptions

Organizing: specialize jobs while coordinating the whole flow

Leading: combine fair incentives with employee voice

Controlling: track output, quality, turnover, and ethical boundaries

Management connection: planning, organizing, leading, and controlling

What Ford changed about management

Ford made product design, production, and staffing one management problem.

High pay could support retention while leaving monotony and exclusion unresolved.

His legacy asks managers to pursue efficiency with adaptation and accountability.

[B] [D] [G] [H]

Questions and discussion

Where should control end?

Which Ford practice would you keep?

Which would you change first, and why?

References: production and strategy

[A] The Henry Ford, “Henry Ford” biography and interview transcript

https://www.thehenryford.org/collections/explore/videos/henry-ford

[B] Library of Congress, “Ford Implements the Moving Assembly Line”

https://guides.loc.gov/this-month-in-business-history/october/ford

[C] Ford Motor Company, “The Model T” (2020)

https://corporate.ford.com/articles/history/the-model-t/

[F] The Henry Ford, “Ford Rouge Factory” research guide

https://askus.thehenryford.org/K12/faq/405833

[G] Bob Casey, “Henry Ford: Case Study of an Innovator” (2021)

https://www.thehenryford.org/collections/explore/articles/henry-ford-case-study-of-an-innovator

[L] The Henry Ford, authenticated Henry Ford quotations, Edison recollections

https://www.thehenryford.org/collections/explore/popular-research-topics/henry-ford-quotations

Sources carried over from the supplied revision; live links not reverified. Full citations also appear in the relevant slide notes.

References: employees and legacy

[D] Matt Anderson, “Ford’s Five-Dollar Day,” The Henry Ford (2014)

https://www.thehenryford.org/collections/explore/articles/fords-five-dollar-day

[E] The Henry Ford, “Henry Ford: Sociological Department”

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[H] U.S. Holocaust Memorial Museum, “Antisemitism and Henry Ford’s ‘The International Jew’”

https://encyclopedia.ushmm.org/content/en/article/antisemitism-and-henry-fords-international-jew

[I] Michigan Supreme Court Learning Center, “Right of Assembly: Battle of the Overpass”

https://www.courts.michigan.gov/49b296/siteassets/msc-learning-center/scrapbook-fordbattleoverpass1937.pdf

[J] Toyota Motor Corporation, “Toyota Production System”

https://global.toyota/en/company/plant-tours/production-system/

[M] The Henry Ford, “Henry Ford and Anti-Semitism”

https://www.thehenryford.org/collections/explore/popular-research-topics/henry-ford-and-anti-semitism

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References: women and the fact check

[N] Detroit Historical Society, “Bennett, Harry”
[O] The Henry Ford, women assembling magnetos, circa 1913 (THF99642)

[P] Ford & Crowther, My Life and Work (1922), chapters VII–VIII
[Q] The New York Times, “For Every Woman, $5 a Day” (Oct. 26, 1916)

[R] The Henry Ford, K–12 Research Guide: Rosie the Riveter
Full URLs and source limitations appear in these slide notes.

Primary accounts reflect their authors’ views; distinguish them from independent evidence.

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Course materials and visual credits

Fundamentals of Management, 11th ed.
Uploaded Pearson chapter slides 1–6, 8, and 9 (2020)

History Module-1.pdf
Uploaded Prentice Hall management-history module (2005)

The Henry Ford digital collections
Historical photographs and documents identified in slide notes